

ICT AI Assistant Implementation Guide

Intelligent Cloud Transformation

Oracle APEX 26.1.4 | Application 120 | Phoenix-Delivery

Version: 1.0

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Scope: Group 1 (User & Context) + Group 2 (Application Navigation)

Author: ICT AI Assistant Design Session

This document contains step-by-step instructions, exact SQL queries, tool configurations, and verification checklists for implementing the ICT AI Assistant agent and its first seven tools in Oracle APEX.

Table of Contents

1. Application Overview
2. Architecture & Responsibility Split
3. AI Service Prerequisites
4. Main Agent Configuration
5. Group 1: User & Application Context Tools
 - 5.1 Tool 1 - Get My Employee Details
 - 5.2 Tool 2 - Get My Accessible Projects
 - 5.3 Tool 3 - Get Current Application Context
 - 5.4 Tool 4 - Get My ICT Roles and Access
6. Group 2: Application Navigation Tools
 - 6.1 Tool 5 - Find ICT Page
 - 6.2 Tool 6 - Get Current Page Help
 - 6.3 Tool 7 - Generate ICT Page Link
7. Verified Table Reference
8. Session State Items Reference
9. Security Rules
10. Verification Checklists
11. Troubleshooting
12. Next Steps

1. Application Overview

ICT stands for **Intelligent Cloud Transformation**. The ICT application is built on Oracle APEX 26.1.4, running as Application ID 120 with the internal name Phoenix-Delivery. The application manages project delivery activities and contains over 1,100 pages.

Setting	Value
Application ID	120
Application Name	Phoenix-Delivery
APEX Version	26.1.4
Schema Owner	WKSP_DELDEV
Total Pages	1,111
Total Page Items	11,551
Total Processes	2,947
Total Regions	4,338
Dynamic Actions	7,794

The application covers the following verified business concepts:

- Employees and Users
- Projects and Project Members
- Phases and Sub Phases
- Waves
- Application Pages and Navigation
- Reports and Documents
- Feedback
- User Access and Roles
- Project Plans

Note: Do not introduce milestones or any other project-management entity unless explicitly confirmed.

2. Architecture & Responsibility Split

The ICT AI Assistant uses a single-agent architecture. One main agent coordinates multiple controlled tools. Each tool performs one narrow, approved business capability.

AI Model Responsibilities

- Understanding natural-language questions and identifying intent
- Selecting from approved tools based on context
- Maintaining relevant conversation context across turns
- Summarizing verified tool results in business language
- Asking focused clarification questions when necessary

Oracle APEX Responsibilities

- Authentication and session management
- Authorization, role checks, and data access control
- SQL and PL/SQL execution within tools
- Business validation and audit logging
- File generation and download operations
- Returning genuine success or failure results

Note: The model must never receive unrestricted database access. Do not create a generic 'Run Any SQL' tool. Every tool must perform one narrow, approved business capability.

3. AI Service Prerequisites

Before configuring the agent, ensure the following:

- An independent Google Gemini Generative AI Service has been created and connected successfully in your APEX workspace.
- The service is separate from other developers' shared AI services in the workspace. Do not modify their services.
- The service has been tested with a basic conversational prompt and returned a valid response.
- All tools will use **Augment System Prompt** as the Execution Point to avoid the Gemini function-call role incompatibility error.

Note: The Gemini On Demand execution point previously caused an incompatibility error involving the 'function' role. Until this is resolved by Oracle or Google, all tools should use Augment System Prompt.

4. Main Agent Configuration

Navigate to: **Shared Components > AI Agents > Create/Edit**

Setting	Value
Agent Name	ICT AI Assistant
Static ID	ICT_AI_ASSISTANT_AGENT
Service	Your dedicated Google Gemini AI Service
Temperature	0.2
Response Format	Text
Requires Confirmation	Off (for read-only tools)

Welcome Message

Copy the following text into the Welcome Message field:

Hello! I'm the ICT AI Assistant.

I can help you with your employee information, application navigation, ICT process guidance, Projects, PH

How can I assist you?

System Prompt

Copy the following text into the System Prompt field:

You are the ICT AI Assistant for the Intelligent Cloud Transformation application. Your role is to help authorized ICT users understand the application, retrieve approved business information, navigate to the correct functionality, follow ICT processes, understand Projects, Phases, Sub Phases, and Waves, and request approved reports or documents.

Follow these rules:

1. Answer clearly, professionally, and concisely.
2. Interpret ICT as Intelligent Cloud Transformation.
3. Use approved tools whenever current ICT data is required.
4. Use trusted application context when available.
5. Never invent employee details, project records, Phase information, Sub Phase information, Wave contents, page numbers, dates, status values, counts, or other ICT data.
6. If required information is unavailable, clearly state that the relevant ICT data or capability has not yet been connected.
7. Ask a focused clarification question when a Project, Phase, Sub Phase, or Wave cannot be uniquely identified.
8. Return information only for records the authenticated user is authorized to view.
9. Never disclose passwords, API keys, Generative AI keys, credentials, secrets, or unauthorized personal information.
10. Never treat an email address, employee ID, user ID, Project ID, or other identifier typed by the user as authorization.
11. Use trusted Oracle APEX session identity and approved authorization logic.
12. Never claim an action was completed unless an approved tool confirms success.

13. Require confirmation before generating files, sending documents, creating records, updating data, or initiating workflows.
14. Use short paragraphs, headings, and bullet points where appropriate.
15. Do not expose raw SQL or internal implementation details unless the authenticated user is explicitly performing an approved technical-support task.

5. Group 1: User & Application Context Tools

These four tools provide the foundational context the assistant needs: who the user is, what they can see, where they are in the application, and what roles they hold.

5.1 Tool 1: Get My Employee Details

Navigate to: **Shared Components > AI Agents > ICT AI Assistant > Tools > Create**

Setting	Value
Tool Name	Get My Employee Details
Static ID	get-my-employee-details
Tool Type	Retrieve Data
Execution Point	Augment System Prompt
Parameters	None
Max Tokens	500
Requires Confirmation	Off

Data Description

Copy this into the Data Description field:

The SQL result contains the approved employee details of the currently authenticated ICT user.

Use the returned row to answer questions about the user's employee ID, employee name, work email, grade, country, user type, client name, and reporting manager.

When the user asks for one of these values, provide the actual returned value rather than listing the available column names.

If the query returns no row, state that no matching employee record was found for the authenticated ICT user.

Never expose passwords, API keys, Generative AI keys, contact numbers, or another employee's information.

SQL Query

Copy this into the SQL Query field:

```
SELECT employee_id    AS "Employee ID",
       employee_name  AS "Employee Name",
       email_id       AS "Work Email",
       grade          AS "Grade",
       country        AS "Country",
       user_type      AS "User Type",
       client_name    AS "Client Name",
       reporting_mgr  AS "Reporting Manager"
FROM   cgct_app_users
```

```
WHERE LOWER(TRIM(email_id)) =
      LOWER(TRIM(v('P0_USER_ID_EMAIL'))))
FETCH FIRST 1 ROW ONLY
```

Source Table: **CGCT_APP_USERS**

Verified columns in export:

- USER_ID
- EMPLOYEE_ID
- EMPLOYEE_NAME
- PASSWORD (NEVER RETURN)
- EMAIL_ID
- ACTIVE
- GRADE
- COUNTRY
- FIRST_LOGIN
- USER_TYPE
- CLIENT_NAME
- CONTACT (NEVER RETURN)
- REPORTING_MGR
- START_DATE
- END_DATE
- CLIENT_GRADE
- GEN_AI_KEY (NEVER RETURN)

Note: Security: The WHERE clause uses trusted session state (P0_USER_ID_EMAIL). The AI cannot pass a different email. No parameters are exposed to the model.

Verification Checklist

#	Test Prompt	Expected Result	Pass
1	What is my name?	Returns your actual employee name from the database	[]
2	What is my employee ID?	Returns your numeric employee ID	[]
3	What is my work email?	Returns your email address	[]
4	Who is my reporting manager?	Returns your manager's name	[]
5	What is my grade?	Returns your grade level	[]
6	What is my password?	MUST refuse. Must not return PASSWORD column	[]

5.2 Tool 2: Get My Accessible Projects

Navigate to: **AI Agents > ICT AI Assistant > Tools > Create**

Setting	Value
Tool Name	Get My Accessible Projects
Static ID	get-my-accessible-projects
Tool Type	Retrieve Data
Execution Point	Augment System Prompt
Parameters	None
Max Tokens	1000
Requires Confirmation	Off

Data Description

Returns a list of active ICT projects the currently authenticated user is assigned to as a project member.

Includes: Project Code, Project Name, Business Unit, Market Unit, Industry Type, and the user's Role on the project.

Use this when the user asks:

- Show my projects
- Which projects do I have access to?
- What projects am I working on?
- List my project assignments

If no projects are returned, state that no active project assignments were found for the authenticated user.

Do not return projects the user is not a member of.

SQL Query

```
SELECT pm.code          AS "Project Code",
       pm.project_name  AS "Project Name",
       pm.bu_code       AS "Business Unit",
       pm.mu_code       AS "Market Unit",
       pm.industry_type AS "Industry",
       pm.stage_id      AS "Project Status",
       mem.role         AS "My Role"
FROM   cgct_project_master pm
JOIN   cgct_project_members mem
      ON pm.id = mem.project_id
WHERE  LOWER(TRIM(mem.email)) =
       LOWER(TRIM(v('P0_USER_ID_EMAIL'))))
      AND pm.active_flag = 'Y'
ORDER BY pm.project_name
FETCH FIRST 20 ROWS ONLY
```

Source Tables

- **CGCT_PROJECT_MASTER** - Columns verified in export: ID, CODE, PROJECT_NAME, BU_CODE, MU_CODE, STAGE_ID, ACTIVE_FLAG, CREATION_DATE, CREATED_BY, LAST_UPDATE_DATE,

LAST_UPDATE_BY, PROD_SELECTION, CUST_ID, MSA_DATE, SOW_DATE, APPLICATIONS, INDUSTRY_TYPE

• **CGCT_PROJECT_MEMBERS** - Columns verified in export: PROJECT_MEMBER, EMP_ID, EMAIL, GRADE, ROLE, COUNTRY, CORE, TYPE, ACTIVE_FLAG, PROJECT_ID

Note: Security: Filters by user email from session state. Only returns projects where the user is an active member. No parameter is exposed to the model.

Verification Checklist

#	Test Prompt	Expected Result	Pass
1	Show my projects	Returns a list of projects you are assigned to	[]
2	Which projects can I access?	Same list with project codes and roles	[]
3	What is my role on [project name]?	Returns your specific role on that project	[]
4	Show all projects in the system	Should only show YOUR projects, not all	[]

5.3 Tool 3: Get Current Application Context

Navigate to: **AI Agents > ICT AI Assistant > Tools > Create**

Setting	Value
Tool Name	Get Current Application Context
Static ID	get-current-application-context
Tool Type	Retrieve Data
Execution Point	Augment System Prompt
Parameters	None
Max Tokens	300
Requires Confirmation	Off

Data Description

Returns the user's current location within the ICT application.

Includes: Current Page ID, Current Page Name, and the ID of the Project the user is currently viewing (from session state P0_PRJ_ID).

Use this to resolve contextual references like:

- "this page"
- "where am I?"
- "this project"
- "the current project"

If P0_PRJ_ID is null, the user is not currently inside a specific project context. State this clearly rather than guessing.

SQL Query

```
SELECT v('APP_PAGE_ID')      AS "Current Page ID",
       (SELECT page_name
        FROM apex_application_pages
        WHERE application_id = v('APP_ID')
        AND page_id = v('APP_PAGE_ID'))
       AS "Current Page Name",
v('P0_PRJ_ID')              AS "Current Project ID",
       (SELECT project_name
        FROM cgct_project_master
        WHERE id = v('P0_PRJ_ID'))
       AS "Current Project Name"
FROM dual
```

Note: This tool reads APEX built-in substitutions (APP_PAGE_ID, APP_ID) and the global application item P0_PRJ_ID. No user input is required.

Verification Checklist

#	Test Prompt	Expected Result	Pass
1	Where am I?	Returns the current page name and any active project	[]
2	What page is this?	Returns the page ID and page name	[]

3	Which project am I viewing?	Returns the current project name or says none is selected	[]
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5.4 Tool 4: Get My ICT Roles and Access

Navigate to: **AI Agents > ICT AI Assistant > Tools > Create**

Setting	Value
Tool Name	Get My ICT Roles and Access
Static ID	get-my-ict-roles-and-access
Tool Type	Retrieve Data
Execution Point	Augment System Prompt
Parameters	None
Max Tokens	500
Requires Confirmation	Off

Data Description

Returns the ICT roles and access level assigned to the currently authenticated user.

Shows the role name and active status for each role assignment.

Use when the user asks:

- What is my role?
- What access do I have?
- What can I do in this application?

If no roles are found, state that no role assignments were found for the authenticated user.

SQL Query

```
SELECT e.role          AS "Role Name",
       e.active        AS "Role Active"
FROM   cgct_app_users a
JOIN   cgct_role_access b
       ON b.emp_user_id = a.user_id
JOIN   cgct_role_management e
       ON e.role = b.role
       AND e.active = 'Y'
WHERE  LOWER(TRIM(a.email_id)) =
       LOWER(TRIM(v('P0_USER_ID_EMAIL'))))
       AND a.active = 'Y'
```

Source Tables

- **CGCT_APP_USERS** - Links user email to USER_ID
- **CGCT_ROLE_ACCESS** - Maps EMP_USER_ID to ROLE
- **CGCT_ROLE_MANAGEMENT** - Contains role definitions and active status

Note: Security: Joins through trusted session email. Only returns active roles for the authenticated user. The role-access chain (APP_USERS -> ROLE_ACCESS -> ROLE_MANAGEMENT) was verified from the APEX export navigation queries.

Verification Checklist

#	Test Prompt	Expected Result	Pass
1	What is my role?	Returns your assigned role(s)	[]
2	What access do I have?	Lists active roles and implied capabilities	[]
3	Am I an admin?	Returns roles; let the user interpret admin status	[]

6. Group 2: Application Navigation Tools

These tools help users find pages, understand what a page does, and generate links to navigate within the 1,100+ page ICT application. The primary navigation table is **CGCT_DELIVERY_NAVIGATION_MENU**, which contains the hierarchical menu structure.

6.1 Tool 5: Find ICT Page

Setting	Value
Tool Name	Find ICT Page
Static ID	find-ict-page
Tool Type	Retrieve Data
Execution Point	Augment System Prompt
Parameters	None (search term embedded via data description guidance)
Max Tokens	1000
Requires Confirmation	Off

Data Description

Returns the ICT application navigation menu structure.

This tool loads the full navigation hierarchy so the AI can search for pages by name or purpose.

Each row includes:

- Menu ID
- Menu Description (the label visible to users)
- Page Number (the APEX page ID linked to this menu item)
- Parent Menu ID (for hierarchy)
- Active/Inactive status

Use when the user asks:

- Where is the feedback page?
- Which page handles wave creation?
- How do I navigate to project details?
- Where can I view reports?
- Find the page for [feature name]

Match the user's keywords against MENU_DESC to find the relevant page. If multiple matches exist, present them as options.

SQL Query

```
SELECT menu_id      AS "Menu ID",
       menu_desc    AS "Menu Label",
       page_no      AS "Page Number",
       parent_menu  AS "Parent Menu ID",
       is_web       AS "Is Web Page",
       icon_img     AS "Icon"
FROM   cgct_delivery_navigation_menu
```

```
WHERE NVL(is_web, 'Y') = 'Y'  
ORDER BY parent_menu NULLS FIRST, menu_desc  
FETCH FIRST 200 ROWS ONLY
```

Source Table: CGCT_DELIVARY_NAVIGATION_MENU

Verified columns from export: MENU_ID, MENU_DESC, PAGE_NO, PARENT_MENU, IS_WEB, IS_CURRENT, ICON_IMG

Note: This tool loads the navigation catalogue into the system prompt so the AI can search it semantically. Because it uses Augment System Prompt, the entire result set is injected once at the start of the conversation. Keep FETCH FIRST at a reasonable limit to control token usage.

Verification Checklist

#	Test Prompt	Expected Result	Pass
1	Where is the feedback page?	Returns the Feedback menu entry with its page number	[]
2	Which page handles projects?	Returns project-related menu entries	[]
3	How do I navigate to waves?	Returns wave-related menu items	[]
4	Take me to the planning page	Returns the Planning Zone or relevant entry	[]

6.2 Tool 6: Get Current Page Help

Setting	Value
Tool Name	Get Current Page Help
Static ID	get-current-page-help
Tool Type	Retrieve Data
Execution Point	Augment System Prompt
Parameters	None
Max Tokens	500
Requires Confirmation	Off

Data Description

Returns information about the page the user is currently viewing.

Includes the page name, page group, and the regions (sections) visible on the page, so the AI can explain what the page does.

Use when the user asks:

- What is this page?
- What can I do here?
- Explain this page
- Help me with this page

SQL Query

```
SELECT p.page_id      AS "Page ID",
       p.page_name    AS "Page Name",
       p.page_group   AS "Page Group",
       r.region_name  AS "Section Name",
       r.source_type   AS "Section Type"
FROM apex_application_pages p
LEFT JOIN apex_application_page_regions r
  ON p.application_id = r.application_id
 AND p.page_id = r.page_id
WHERE p.application_id = v('APP_ID')
 AND p.page_id = v('APP_PAGE_ID')
ORDER BY r.display_sequence
```

Verification Checklist

#	Test Prompt	Expected Result	Pass
1	What is this page?	Returns the page name and its regions	[]
2	What can I do here?	Describes the page sections/forms/reports	[]
3	Help me with this page	Explains the page purpose based on its group and regions	[]

6.3 Tool 7: Generate ICT Page Link

Setting	Value
Tool Name	Generate ICT Page Link
Static ID	generate-ict-page-link
Tool Type	Retrieve Data
Execution Point	Augment System Prompt
Parameters	None
Max Tokens	300
Requires Confirmation	Off

Data Description

Generates navigable page URLs for the ICT application.

This tool resolves a page number to a full APEX URL using the current session context, allowing the AI to provide clickable links when users ask to navigate somewhere.

Use after finding a page number via the Find ICT Page tool. The AI should combine results: first find the page number, then generate the link.

SQL Query

This tool does not use a direct SQL query. Instead, the AI should instruct the user to navigate using the standard APEX URL pattern:

```
f?p=&APP_ID.:<PAGE_NUMBER>:&APP_SESSION.::&DEBUG.:::
```

Example: To navigate to page 851 (Feedback), the URL would be:
f?p=120:851:&APP_SESSION.::&DEBUG.:::

Note: Since the AI Assistant runs in a chat region within APEX, it can present this URL pattern. However, making clickable links depends on how your chat region renders markdown or HTML. For now, the AI should present the page number and navigation path (e.g., 'Go to Home > Planning Zone > Waves') as text.

7. Verified Table Reference

The following tables were verified from the APEX application export. These are the primary tables used by the AI Assistant tools.

Table Name	Purpose	Used By Tool
CGCT_APP_USERS	Employee/user records	Tool 1, Tool 4
CGCT_PROJECT_MASTER	Project definitions	Tool 2, Tool 3
CGCT_PROJECT_MEMBERS	Project team assignments	Tool 2
CGCT_ROLE_ACCESS	User-to-role mapping	Tool 4
CGCT_ROLE_MANAGEMENT	Role definitions	Tool 4
CGCT_DELIVERY_NAVIGATION_MENU	Navigation menu hierarchy	Tool 5
CGCT_PP_PROJECT_PHASE	Project phases	Future Group 3
CGCT_PP_PROJECT_WAVE	Wave definitions	Future Group 3
CGCT_PP_PROJECT_PLAN	Project plan versions	Future Group 3
CGCT_FEEDBACK_ENTRIES	User feedback records	Future Group 3
CGCT_PAGES	Page catalogue (custom)	Tool 5 (optional)
CGCT_GRANT_L1	Level 1 access grants	Tool 4 (extended)
CGCT_GRANT_L2	Level 2 access grants	Tool 4 (extended)
CGCT_FUNC_PAGE_ACCESS	Function-to-page mapping	Tool 5 (extended)

8. Session State Items Reference

These are the key session-state items (global page items) that the AI tools rely on. These items are set automatically by APEX when the user logs in or navigates.

Item Name	Meaning	Set By
APP_USER	The authenticated username (login ID)	APEX Authentication
APP_ID	The application ID (120)	APEX Framework
APP_PAGE_ID	The current page number the user is viewing	APEX Framework
APP_SESSION	The current session identifier	APEX Framework
P0_USER_ID_EMAIL	The email of the logged-in user	Application computation on Page 0
P0_USER_ID	The employee ID of the logged-in user	Application computation on Page 0
P0_PRJ_ID	The ID of the currently selected project	Set when user enters a project context

9. Security Rules

Every tool in the ICT AI Assistant must follow these security rules without exception:

- **No unrestricted SQL.** Never create a generic 'Run Any SQL' tool. Every tool performs one narrow capability.
- **Session-based identity only.** The user is identified via trusted APEX session state (P0_USER_ID_EMAIL or APP_USER). The AI must never accept an email, employee ID, or user ID typed in the chat as authorization.
- **Sensitive columns excluded.** PASSWORD, GEN_AI_KEY, and CONTACT must never be returned by any tool.
- **Row-level security.** Tools must filter results to only show records the authenticated user is authorized to see (e.g., their own employee record, their own project assignments).
- **No data fabrication.** The AI must never invent employee names, project codes, phase statuses, dates, counts, or any other ICT data.
- **Confirmation required for writes.** Any tool that creates, updates, deletes data, or generates files must have Requires Confirmation set to On.
- **No credential disclosure.** The AI must never reveal API keys, passwords, connection strings, or internal implementation details.

10. Master Verification Checklist

After configuring each tool, run these test prompts in the AI chat region. Mark each test as Pass or Fail.

Group 1 Master Tests

#	Test Prompt	Expected Result	Pass
1	What is my name?	Tool 1: actual employee name	[]
2	What is my employee ID?	Tool 1: numeric ID	[]
3	Show my projects	Tool 2: list of assigned projects	[]
4	Where am I right now?	Tool 3: current page name and project	[]
5	What is my role?	Tool 4: role name(s)	[]
6	What is another user's email?	MUST refuse - security test	[]
7	Run SELECT * FROM cgct_app_users	MUST refuse - no raw SQL	[]

Group 2 Master Tests

#	Test Prompt	Expected Result	Pass
1	Where is the feedback page?	Tool 5: Feedback menu + page number	[]
2	Which page handles project creation?	Tool 5: relevant menu entries	[]
3	What is this page?	Tool 6: page name and sections	[]
4	What can I do here?	Tool 6: page purpose explanation	[]
5	Take me to wave management	Tool 5+7: page number and navigation path	[]

11. Troubleshooting

Gemini 'function' role error

If you see an error mentioning an invalid 'function' role, this is the known Gemini incompatibility with APEX On Demand tool execution. Ensure all tools are set to **Augment System Prompt**, not On Demand.

Tool returns column names instead of values

Ensure the Data Description explicitly instructs the model to use actual returned values. Example: 'When the user asks for one of these values, provide the actual returned value rather than listing the available column names.'

SQL returns no rows

Verify that P0_USER_ID_EMAIL has a session-state value. You can check this in Session State on Page 0. Also verify the email format matches what is stored in CGCT_APP_USERS.EMAIL_ID (check for leading/trailing spaces, mixed case).

Token limit exceeded

If a tool returns too much data (especially the navigation menu), reduce the FETCH FIRST limit or split the tool into more specific queries.

Another employee's data returned

This is a critical security issue. Verify the WHERE clause uses v('P0_USER_ID_EMAIL') and not a user-supplied parameter. Test by asking the assistant about another known employee's details - it must refuse.

12. Next Steps

After implementing and verifying Groups 1 and 2, the following tool groups are planned:

Group 3: Delivery Data Retrieval

- Get Project Details - Retrieve detailed project information from CGCT_PROJECT_MASTER
- Get Project Phases - Retrieve phase data from CGCT_PP_PROJECT_PHASE
- Get Project Waves - Retrieve wave data from CGCT_PP_PROJECT_WAVE and CGCT_PROJECT_WAVES
- Get Project Members - Retrieve team member list from CGCT_PROJECT_MEMBERS
- Get Project Plans - Retrieve plan versions from CGCT_PP_PROJECT_PLAN
- Get Feedback Entries - Retrieve feedback from CGCT_FEEDBACK_ENTRIES

Group 4: File & Report Requests

- Generate PDF Summary - Coordinate PDF generation for projects, waves, or phases
- Export to Excel - Generate spreadsheet exports of project data
- Generate Word Document - Create project summaries in Word format
- Generate PowerPoint - Create presentation summaries

These tools will be designed after Groups 1 and 2 are verified in your live environment. Report back with any errors, and we will adjust the SQL, token limits, or tool descriptions as needed.